Illustration 1.11 Consider the functions:

$$\text{(i) } f(x) = x^2 \quad \text{(ii) } f(x) = x^2 + 1 \quad \text{(iii) } f(x) = (x+1)^2.$$

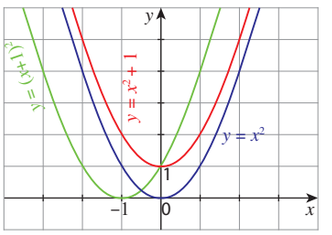


Figure 1.54

$f(x) = x^2 + 1$ causes the graph of the function $f(x) = x^2$ shifts to the upward for one unit.

$f(x) = (x+1)^2$ causes the graph of the function $f(x) = x^2$ shifts to the left for one unit. (See Figure 1.54.)

Illustration 1.12 Compare and contrast the graphs $y = x^2 - 1, y = 4(x^2 - 1)$ and $y = (4x)^2 - 1$

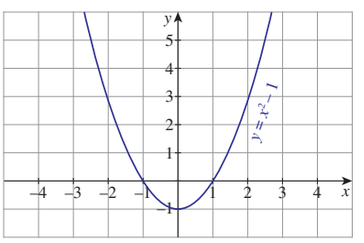


Figure 1.55

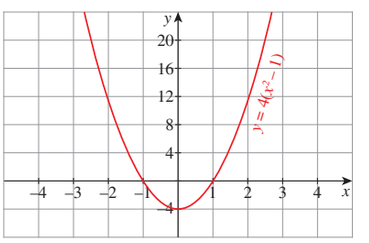


Figure 1.56

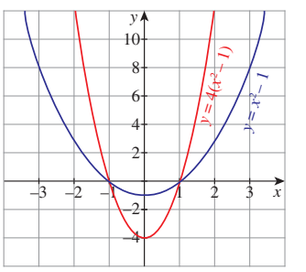


Figure 1.57

The graphs Figures 1.55 and 1.56 look identical until we compare the scales on the y-axis. The scale in Figure 1.56 is four times as large, reflecting the multiplication of the original function by 4 (Figure 1.55). The effect looks different when the functions are plotted on the same scale as in Figure 1.57.

The graph of $y = (4x)^2 - 1$ is shown in Figure 1.58. Can you spot the difference between Figure 1.55 and Figure 1.58? In this case, x-scale has now changed, by the same factor of 4 as in the function (Figure 1.58). To see this, note that substituting $x = \frac{1}{4}$ into $(4x)^2 - 1$ produces $1^2 - 1$, exactly the same as substituting $x = 1$ into the original function (Figure 1.55). When plotted on the same set of axes (as in Figure 1.59) the parabola $y = (4x)^2 - 1$ looks thinner. Here, the x-intercepts are different, but y-intercepts are the same.

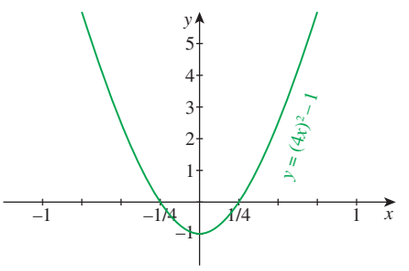


Figure 1.58

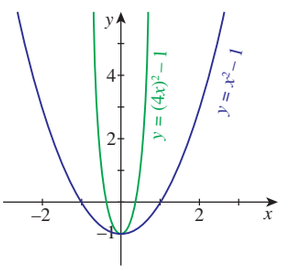


Figure 1.59